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Electrolytic capacitor

Abstract

Provided is an electrolytic capacitor with a resin layer, in which an increase in ESR over time is suppressed. A electrolytic capacitor includes a capacitor element including an anode foil, a cathode foil, and electrolytic solution, a case housing the capacitor element, a sealing member sealing the case, and a resin layer arranged in the vicinity of the sealing member. The resin layer arranged in the vicinity of the sealing member includes epoxy resin composition without ester bond.

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Background/Summary

FIELD OF INVENTION

(1) The present disclosure relates to an electrolytic capacitor having a resin layer.

BACKGROUND

(2) In electrolytic capacitors utilizing valve action metal such as tantalum and aluminum, etc., by making the valve action metal as the anode electrode into shapes of a sintered body or an etching foil, etc., to enlarge the surface of dielectric, the electrolytic capacitors can be in a small size and have a large capacity. This type of electrolytic capacitor is formed by filling voids with an electrolyte so that the dielectric oxide film of the anode electrode is brought into close contact with the counter electrode. That is, the electrolytic capacitors include liquid electrolytic capacitors having only an electrolytic solution, hybrid electrolytic capacitors having an electrolytic solution and a solid electrolyte, and bipolar electrolytic capacitors having a dielectric film formed on both electrodes.

(3) The electrolytic solution contains ethylene glycol or γ -butyrolactone as a solvent, and contains a carboxylic acid such as 1,6-decandicarboxylic acid, 1,7-octanedicarboxylic acid, or azelaic acid or a salt thereof as a solute. This electrolytic solution comes into direct contact with the dielectric film and acts as a true cathode electrode, and also has a repairing action on the dielectric film. However, evaporation and volatilization occurs, where the electrolytic solution is released outside the electrolytic capacitor over time. Therefore, the capacitance of the electrolytic capacitor decreases over time toward dry-up, and the tangent ($\tan \delta$) of the loss angle increases over time, and finally reaching the end of its life.

(4) Accordingly, a capacitor element is inserted into an outer case with bottom, and the opening of the outer case is sealed with a sealing member to seal the electrolytic solution impregnated in the capacitor element, thereby suppressing evaporation and volatilization of the electrolytic solution and extending its life. However, elastomers such as butyl rubber and ethylene propylene diene rubber (EPDM) are used for the sealing member. Therefore, the electrolytic solution is not completely confined within the case, but permeates the sealing member and gradually volatilizes to the outside of the electrolytic capacitor. Therefore, a proposal has been proposed to cover the sealing member with a resin layer to suppress volatilization of the electrolytic solution by permeation through the sealing member, as shown in Patent Document 1.

(5) Resin layers generally have excellent thermal insulation properties. When the thermal insulation property of the electrolytic capacitor is increased, it becomes difficult to dissipate the heat generated in the case, and the heat is easily accumulated. Therefore, the use of ethylene glycol as a solvent of the electrolytic solution has been proposed, as shown in Patent Document 2. This is because the thermal conductivity of ethylene glycol is about twice as large as that of γ -butyrolactone or sulfolane. Although Patent Document 2 suggests that the larger the content ratio of ethylene glycol in the solvent, the higher the thermal conductivity, it is said that the ethylene glycol is preferably 90 mass % or less in order to contain other types of solvents.

PRIOR ART DOCUMENT

Patent Document

(6) Patent Document 1: JP S60-245106 A

(7) Patent Document 2: WO2018/123525

SUMMARY OF INVENTION

Problems to be Solvedb Invention

(8) The resin layer contains a curable resin composition. Examples of the curable resin composition include an epoxy resin compositions containing an epoxy resin and an acid anhydride curing agent. Epoxy resin compositions are compositions of epoxy resins cured with an acid anhydride-base curing agents. The epoxy resin composition containing the acid anhydride curing agent has improved heat resistance, and since the reaction temperature between the epoxy resin and the acid anhydride curing agent is 100° C. or higher, the curing reaction can be accelerated by using the temperature as a trigger. Therefore, it is considered that the epoxy resin composition containing the epoxy resin and the acid anhydride curing agent is suitable for the resin layer.

(9) The present inventors form a part of the electrolytic capacitor with a resin layer with the epoxy resin cured with the acid anhydride curing agent. Then, this electrolytic capacitor was exposed to a temperature environment of 150° C. Then, it was confirmed that the ESR of the electrolytic capacitor exposed to this temperature environment for 400 hours was significantly increased.

(10) The present disclosure has been proposed to solve the above problems, an object of the present disclosure is to provide an electrolytic capacitor with a resin layer, in which an increase in ESR over time is suppressed.

Solution to the Problems

(11) As a result of their diligent research, the present inventors have found that the increase in ESR over time in the electrolytic capacitor is remarkable when an epoxy resin composition containing an epoxy resin and an acid anhydride curing agent (hereinafter referred to as acid anhydride-cured epoxy resin) is included and ethylene glycol is used as the solvent in the electrolytic solution.

(12) As a result of a diligent research based on this knowledge, the present inventors have found since the acid anhydride-cured epoxy resin reacts with water or ethylene glycol, decomposes, and is eluted into the electrolytic solution, the presence of the acid anhydride-cured epoxy resin in the electrolytic solution was the cause of the increase in ESR of the electrolytic capacitor.

(13) Here, the acid anhydride-cured epoxy resins contain ester bonds in their chemical structure. On the other hand, even when water is not intentionally added to the electrolytic capacitor, a small amount of water is mixed into the electrolytic capacitor during the manufacturing process. Ethylene glycol is a compound with a hydroxy group, and the compound with a hydroxy group has a structure that is easily accessible to water. Under these circumstances, when the compound with a hydroxy group permeates the sealing member, water also permeates with it, and the ester bond of the acid anhydride-cured epoxy resin reacts with water, and the acid anhydride-cured epoxy resin is hydrolyzed. In addition, an esterification reaction between a carbonyl group produced by hydrolysis and a compound with a hydroxy group also occurs. As described above, it is considered that a part of the acid anhydride-cured epoxy resin component that reacted with water or the compound with a hydroxy group permeated the sealing member and eluted into the electrolytic solution.

(14) The present disclosure has been made based on these finding, and a electrolytic capacitor of the present disclosure includes a capacitor element including an anode foil, a cathode foil, and electrolytic solution, a case housing the capacitor element, a sealing member sealing the case, and a resin layer arranged in the vicinity of the sealing member, in which the resin layer include epoxy resin composition without ester bond.

(15) The epoxy resin composition may be made from a phenol-based curing agent or an amine-based curing agent and an epoxy resin as raw materials. The epoxy resin composition may have a chemical structure containing a phenol-based curing agent or an amine-based curing agent and an epoxy resin.

(16) The electrolytic solution may contain at least one of a compound with hydroxy group, sulfolane and γ -butyrolactone.

(17) The compound with hydroxy group may be at least one of ethylene glycol, diethylene glycol, or polyethylene glycol.

(18) The capacitor element may further comprise a solid electrolyte.

(19) Cationic component in the electrolytic solution may be contained in an amount of 76 mmol or less per 100 g of the electrolytic solution. Cationic component in the electrolytic solution may be contained in an amount of 25 mmol or less per 100 g of the electrolytic solution.

Effects of Disclosure

(20) According to the present disclosure, an electrolytic capacitor with a resin layer, in which an increase in ESR over time is suppressed can be provided.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a fluoroscopic image taken from multiple directions of Comparative Example 1, Example 1, and Example 2.

DETAILED DESCRIPTION

(2) Hereinafter, the electrolytic capacitor according to the embodiment of the present disclosure will be described. The present disclosure is not limited to the following examples.

(3) (Overall Configuration)

(4) The electrolytic capacitor is a passive element that stores and discharges electric charges according to the capacitance. These electrolytic capacitors include liquid electrolytic capacitors having only an electrolytic solution, and hybrid electrolytic capacitors in which a solid electrolyte such as a conductive polymer or gel and an electrolytic solution are used in combination. These electrolytic capacitors include electrolytic capacitors with dielectric oxide film intentionally formed only on the anode electrode side and bipolar electrolytic capacitors with dielectric oxide film formed on both electrodes. Furthermore, these electrolytic capacitors include electrolytic capacitors with lead terminals drawn out in appearance, electrolytic capacitors that are surface-mounted on a board with a base, and electrolytic capacitors that are entirely covered with resin along with other electronic circuit components. These various combinations of electrolytic capacitors are hereinafter collectively referred to simply as electrolytic capacitors.

(5) Electrolytic capacitors have a capacitor element, a case, and a sealing member. The case houses the capacitor element. The sealing member is attached to an opening of the case by a crimping process to seal the opening of the case. The capacitor element has an anode foil, a cathode foil, a separator, and electrolytic solution. The anode foil and the cathode foil face each other via a separator. A dielectric oxide film layer is formed on the anode foil. A dielectric oxide film layer is also formed on the cathode foil, if necessary. In addition to the electrolytic solution, solid electrolytes may be included. The solid electrolyte is interposed between the anode foil and the cathode foil, and is in close contact with the dielectric oxide film layer. The electrolytic solution fills voids of the capacitor element. Electrolytic capacitors may also have a base if necessary.

(6) (Sealing Member)

(7) The sealing member is attached to the case by a crimping process. The case is made of aluminum, aluminum alloy containing aluminum or manganese, or stainless steel, and is a cylinder, for example, with a bottom and an opening at the other end. In the crimping process, the opening of the case is bent inward and collapsed to tightly fit the case to the sealing member. The sealing member with elastic force is a plate body made of an elastomer, or a plate body in which a synthetic resin plate or a metal plate is laminated on the elastomer. The elastomer adheres closely, by elastic force, to the case which the crimping process has been performed, and maintains airtightness in the case. However, permeability to volatilized electrolytic solution of sealing members containing elastomer is not zero. These elastomers include butyl rubber, ethylene propylene diene rubber (EPDM), silicone rubber, and butyl rubber.

(8) (Resin Layer)

(9) The electrolytic capacitor further includes a resin layer. The resin layer suppresses the transpiration of the electrolytic solution. That is, it may be provided at any location of the electrolytic capacitor as long as it is in the vicinity of the sealing member through which the electrolytic solution may pass. For example, the resin layer covers at least a partial area of the surface of the sealing member. This resin layer covers the sealing member by resin molding, and reduces the permeability through which the electrolytic solution permeates the sealing member. That is, vicinity does not have to be spaced apart and includes contact arrangements. Further, for example, when the electrolytic capacitor is a chip type provided with a base, the resin layer is provided, for example, at or around a place where the base and the case come into contact with each other and near the sealing member. Transpiration of the electrolytic solution that permeates the sealing body is suppressed by this resin layer. In other words, this resin layer may also react with water or ethylene glycol in the electrolytic solution that has passed through the sealing body and elute into the electrolytic solution through the sealing body.

(10) This resin layer does not use the acid anhydride-cured epoxy resin, but is made of a phenol-based cured epoxy resin or an amine-based cured epoxy resin. This resin layer may contain a compound other than the phenol-based cured epoxy resin or the amine-based cured epoxy resin.

(11) The phenol-based cured epoxy resin is an epoxy resin cured with a phenol-based curing agent, and is an epoxy resin composition having the phenol-based curing agent and the epoxy resin as raw materials and containing the phenol-based curing agent and the epoxy resin in the chemical structure. The amine-based cured epoxy resin is an epoxy resin cured with an amine-based curing agent, and is an epoxy resin composition having the amine-based curing agent and the epoxy resin as raw materials and containing the amine-based curing agent and the epoxy resin in the chemical structure. The acid anhydride-cured epoxy resin has an ester bond in the chemical structure, whereas the acid anhydride-cured epoxy resin of the phenol-based cured epoxy resin and the amine-based cured epoxy resin has no ester bond in the chemical structure.

(12) Specifically, the phenol-based cured epoxy resin is produced by the reaction and bonding of the phenolic hydroxy group and the epoxy group of the phenol-based curing agent. In addition, the amine-based cured epoxy resin is produced by the reaction and bonding of the amino group and the epoxy group of the amine-based curing agent. The bonds between these curing agents and the epoxy resin are ether bonds, and there is no risk of hydrolysis.

(13) As can be seen from this reaction system, the epoxy resin composition containing the phenol-based curing agent and the epoxy resin in the chemical structure does not mean the epoxy resin composition incorporated into the chemical structure in the same state as before the reaction. The inclusion of the phenol-based curing agent and the epoxy resin in the chemical structure means that each group is substituted for binding and then incorporated into the chemical structure. In addition, as can be seen from this reaction system, the epoxy resin composition containing the amine-based curing agent and the epoxy resin in the chemical structure does not mean the epoxy resin composition incorporated into the chemical structure in the same state as before the reaction. The inclusion of the amine-based curing agent and the epoxy resin in the chemical structure means that each group is substituted for binding and then incorporated into the chemical structure.

(14) The epoxy resin contained in the phenol-based cured epoxy resin and the amine-based cured epoxy resin is an epoxy oligomer having two or more reactive epoxy group at the ends. This epoxy resin transforms from a liquid substance to a solid resin by bridging between the epoxy resins through the addition reaction of the acid anhydride curing agent. Epoxy resins typically include bisphenol-A diglycidyl ether, which is a condensate of bisphenol A and epichlorohydrin. Examples of the epoxy resin include other glycidyl-type epoxy resins and alicyclic epoxisides such as 3,4-epoxycyclohexylmethyl-3', 4'-epoxycyclohexanecarboxylate.

(15) Examples of the glycidyl-type epoxy resin include bisphenol type obtained by glycidylating bisphenols. Examples of bisphenols include bisphenol A, bisphenol F, bisphenol AD, bisphenol S, tetramethylbisphenol A, tetramethylbisphenol F, tetramethylbisphenol AD, tetramethylbisphenol S,

tetrabromobisphenol A, tetrachlorobisphenol A, and tetrafluorobisphenol A.

(16) In addition, examples of the glycidyl-type epoxy resin include epoxy resin obtained by glycidylating divalent phenols. Examples of divalent phenols include biphenol, dihydroxynaphthalene, and 9,9-bis(4-hydroxyphenyl)fluorene, and the like.

(17) In addition, examples of the glycidyl-type epoxy resin include epoxy resin obtained by glycidylating triphenols. Examples of triphenols include 1,1,1-tris(4-hydroxyphenyl)methane and 4,4-(1-(4-(1-(4-hydroxyphenyl)-1-methylethyl)phenyl)ethylidene)bisphenol, and the like.

(18) In addition, examples of the glycidyl-type epoxy resin include epoxy resin obtained by glycidylating tetrakisphenols. Examples of tetrakisphenols include 1,1,2,2-tetrakis(4-hydroxyphenyl)ethane, and the like.

(19) In addition, examples of the glycidyl-type epoxy resin include novolak-type epoxy resin obtained by glycidylating novolaks. Examples of the novolaks include phenol novolak, cresol novolak, bisphenol A novolak, brominated phenol novolak, brominated bisphenol A novolak, and the like.

(20) In addition, examples of the glycidyl-type epoxy resin include epoxy resin obtained by glycidylating polyhydric phenols, and aliphatic-ether-type epoxy resin obtained by glycidylating polyhydric alcohols such as glycerin and polyethylene glycol.

(21) In addition, examples of the glycidyl-type epoxy resin include ether-ester-type epoxy resin obtained by glycidylating hydroxy carboxylic acid, ester-type epoxy resin obtained by glycidylating polycarboxylic acid, glycidylated amine compounds, and amine-type epoxy resin. Examples of hydroxy carboxylic acids include p-oxybenzoic acid and β -oxynaphthoic acid, and the like. Examples of polycarboxylic acids include phthalic acid, terephthalic acid, and the like. Examples of amine compounds include 4,4-diaminodiphenylmethane and m-aminophenol. Examples of the amine-type epoxy resin include triglycidyl isocyanurate, and the like.

(22) Examples of the phenol-based curing agent contained in the phenol-based cured epoxy resin include bifunctional phenols and polyfunctional phenols, and the like. Examples of the bifunctional phenols include hydroquinone, resorcinol, bisphenol F, biphenol, tetrabromobisphenol A and naphthalene diol, and the like. Examples of the polyfunctional phenols include phenol novolak resins, and the like.

(23) Examples of the amine-based curing agent contained in the amine-based cured epoxy resin include aliphatic polyamines, aromatic polyamines, and modified amines, and the like. Examples of the aliphatic polyamine include diethylenetriamine and triethylenetetramine, and the like. Examples of the aromatic polyamine include meta-phenylenediamine, diaminodiphenylmethane, diaminodiphenyl sulfone and the like. Examples of the modified amine include amine adduct and ketimine, and the like.

(24) (Capacitor Element)

(25) (Electrolytic Solution)

(26) The electrolytic solution impregnated in the capacitor element is, for example, a solution of an ion dissociative salt that dissociates into an anionic component and a cationic component, or a solvent that does not contain an ion dissociative salt. Examples of the solvent include compounds with hydroxy group, cyclic lactones, and sulfone compounds. Examples of compounds with hydroxy group include protic organic polar solvents. Examples of the protic organic polar solvents include monohydric alcohols, polyhydric alcohols, and oxyalcohol compounds. Examples of the monohydric alcohols include ethanol, propanol, butanol, pentanol, hexanol, cyclobutanol, cyclopentanol, cyclohexanol, benzyl alcohol, and the like. Examples of the polyhydric alcohols and the oxyalcohol compounds include ethylene glycol, diethylene glycol, propylene glycol, glycerin, methyl cellosolve, ethyl cellosolve, methoxypropylene glycol, dimethoxypropanol, alkylene oxide adducts of polyhydric alcohols such as polyethylene glycol and polyoxyethylene glycerin, and the like. Examples of the cyclic lactones include γ -butyrolactone, γ -valerolactone, δ -valerolactone, and the like. Examples of the sulfone compound include chain sulfone and cyclic sulfone. Examples of

the chain sulfone include, for example, dimethyl sulfone, diethyl sulfone, dipropyl sulfone, and diphenyl sulfone. Examples of the cyclic sulfone include, for example, sulfolane, 3-methylsulfolane, and 2,4-dimethylsulfolane.

(27) Here, the resin layer included in the sealing member is the phenol-based cured epoxy resin or the amine-based cured epoxy resin. The phenol-based cured epoxy resin and the amine-based cured epoxy resin do not have ester bonds. Therefore, even if the solvent of the electrolytic solution contains the compound with hydroxy group, the phenol-based cured epoxy resin and the amine-based cured epoxy resin are not decomposed by the compound with hydroxy group. Therefore, even if the solvent of the electrolytic solution contains the compound with hydroxy group, when the resin layer provided in the sealing member is the phenol-based cured epoxy resin or the amine-based cured epoxy resin, the resin component does not elute into the electrolytic solution, and thereby suppressing the increase in ESR of the electrolytic capacitor over time.

(28) When the solid electrolyte is used in combination with the electrolytic solution containing ethylene glycol as a solvent, the conductive polymer's electrical conductivity is improved by a change in the higher-order structure of the conductive polymer and reorientation of the crystal structure of the polymer chain. Cyclic lactones improve ESR properties at low temperatures. Since the sulfone compound has a high boiling point, it suppresses the transpiration of the electrolytic solution and improves the high temperature characteristics. Ethylene glycol is particularly preferable as the solvent.

(29) From a viewpoint of suppressing the leakage of the electrolytic solution from the electrolytic capacitor through the resin layer, Ethylene glycol is preferable as the solvent of the electrolytic solution, and then sulfolane is preferable. A mixed solvent of ethylene glycol and sulfolane may be used for the electrolytic solution.

(30) Examples of organic acids that become anionic components as solute include oxalic acid, succinic acid, glutaric acid, pimelic acid, suberic acid, sebacic acid, phthalic acid, isophthalic acid, terephthalic acid, maleic acid, adipic acid, benzoic acid, toluylic acid, enanthic acids, malonic acids, carboxylic acids such as 1,6-decandicarboxylic acid, 1,7-octanedicarboxylic acid, azelaic acid, resorcinic acid, fluorochloric acid, gallic acid, gentisic acid, protocatechuic acid, pyrocatechuic acid, trimellitic acid, and pyromellitic acid, phenols, and sulfonic acids. Examples of the inorganic acid include boric acid, phosphoric acid, phosphorus acid, hypophosphorous acid, carbonic acid, silicic acid, and the like. Examples of complex compounds of organic and inorganic acids include borodisalicylic acid, borodioxalic acid, borodiglycolic acid, borodimalonic acid, borodichuicinic acid, borodiadipic acid, borodiazelaic acid, borodibenzoic acid, borodimarainic acid, borodilactic acid, borodiapple acid, borodi tartric acid, borodicitrate acid, borodiphthalic acid, borodi (2-hydroxy) isobutyric acid, borodiresorcinic acid, borodimethylsalicylic acid, borodinaftoeic acid, borodimandelic acid and borodi (3-hydroxy) propionic acid, and the like.

(31) Further, examples of at least one salt of the organic acid, the inorganic acid, and the complex compound of the organic acid and the inorganic acid include ammonium salts, quaternary ammonium salts, quaternary amidinium salts, amine salts, sodium salts, and potassium salts, and the like. Examples of the quaternary ammonium ion of the quaternary ammonium salts include tetramethylammonium, triethylmethylammonium, tetraethylammonium, and the like. Examples of the quaternary amidinium salts include ethyldimethylimidazolinium and tetramethylimidazolinium, and the like. Examples of the amine salts include primary amines, secondary amines, and tertiary amines. Examples of the primary amines include methylamine, ethylamine, propylamine, and the like, examples of the secondary amines include dimethylamine, diethylamine, ethylmethylamine and dibutylamine, and the like, and examples of the tertiary amines include trimethylamine, triethylamine, tributylamine, ethyldimethylamine, and ethyldiisopropylamine, and the like.

(32) The cationic component of the solute is preferably added in an amount of 76 mmol or less per 100 g of the electrolytic solution, further preferably 51 mmol or less per 100 g of the electrolytic solution, more preferably 34 mmol or less per 100 g of the electrolytic solution, particularly

preferably 25 mmol or less per 100 g of the electrolytic solution, and most preferably 17 mmol or less per 100 g of the electrolytic solution. The ratio of the cationic and anionic components of the solute may be equimolecular amount, or there may be an excess of the cationic component or an excess of the anionic component.

(33) The cationic component tends to promote the dedoping reaction of the conductive polymer in the solid electrolyte in a high thermal environment such as a reflow process when mounting the electrolytic capacitor. When the dedoping reaction is promoted by the cationic component, the electrical conductivity of the solid electrolyte decreases and the ESR of the electrolytic capacitor increases. On the other hand, when the resin layer is arranged in the vicinity of the sealing member, the resin layer inhibits the transpiration of the cationic component through the sealing member, and the decrease of the cationic component in the electrolytic capacitor is suppressed. Therefore, when the resin layer is arranged in the vicinity of the sealing member, dedoping reactions due to the cationic component occur frequently, and the ESR of the electrolytic capacitor increases.

(34) However, if the amount of the cationic component of the solute in the electrolytic solution is 76 mmol or less per 100 g of the electrolytic solution, the increase in ESR is suppressed as compared with the case where the amount exceeds 76 mmol. Furthermore, if the amount of the cationic component of the solute in the electrolytic solution is 51 mmol or less per 100 g of the electrolytic solution, the increase in ESR is dramatically reduced compared with the case where the amount is 76 mmol per 100 g of the electrolytic solution. If the amount of the cationic component of the solute in the electrolytic solution is 25 mmol or less per 100 g of the electrolytic solution, the increase in ESR is further dramatically reduced compared with the case where the amount is 51 mmol per 100 g of the electrolytic solution.

(35) Further, other additives may be added to the liquid. Examples of additives include complex compounds of boric acid and polysaccharides (mannit, sorbit, etc.), complex compounds of boric acid and polyhydric alcohol, borate esters, nitro compounds (o-nitrobenzoic acid, m-nitrobenzoic acid, p-nitrobenzoic acid, o-nitrophenol, m-nitrophenol, p-nitrophenol, p-nitrobenzylalcohol etc.), phosphate esters, and the like. These may be used alone, or two or more kinds may be combined.

(36) (Electrode Foil)

(37) The anode foil and the cathode foil is a long foil body made of a valve acting metal. The valve acting metal is aluminum, tantalum, niobium, niobium oxide, titanium, hafnium, zirconium, zinc, tungsten, bismuth, antimony and the like. The purity is preferably 99.9% or more for the anode foil and is preferably about 99% or more for the cathode foil, however impurities such as silicon, iron, copper, magnesium, and zinc may be contained.

(38) The surface of the anode foil is enlarged as a sintered body obtained by sintering powder of a valve action metal or an etching foil obtained by etching a stretched foil. The enlarged surface structure is formed with tunnel-shaped etching pits, sponge-shaped etching pits, or voids between dense powders. Typically, the enlarged surface structure is formed by direct current etching or alternating current etching in which direct current or alternating current is applied in an acidic aqueous solution containing halogen ions such as hydrochloric acid, or is formed by depositing or sintering metal particles or the like on the core. The cathode foil may also have the enlarged surface structure by deposition, sintering or etching.

(39) A dielectric oxide film is typically an oxide film formed on the surface layer of the anode foil, and when the anode foil is made of aluminum, it is an aluminum oxide obtained by oxidizing a porous structural region. This dielectric oxide film is formed by a chemical conversion treatment in which a voltage is applied in an aqueous solution of adipic acid, boric acid or phosphoric acid, and the like. Further, a thin dielectric oxide film (about 1 to 10 V) may be formed on the surface layer of the cathode foil by chemical conversion treatment, if necessary. In addition, the dielectric oxide film may be formed by vapor deposition of a layer consisting of metal nitride, metal carbide, or metal carbonitride, or may be produced by using material containing carbon on the surface.

(40) (Separator)

(41) Separator includes celluloses such as kraft, Manila hemp, esparto, hemp, and rayon, and mixed papers thereof, polyester resins such as polyethylene terephthalates, polybutylene terephthalates, polyethylene naphthalates, and derivatives thereof, polytetrafluoroethylene resins, polyvinylidene fluoride resin, vinylon resin, aliphatic polyamide, semi-aromatic polyamide, polyamide resin such as total aromatic polyamide, polyimide resin, polyethylene resin, polypropylene resin, trimethylpentene resin, polyphenylene sulfide resin, acrylic resin, polyvinyl alcohol resin and the like, these resins may be used alone or in combination.

(42) (Solid Electrolyte)

(43) When the solid electrolyte is formed in the capacitor element, the solid electrolyte contains a conductive polymer. The conductive polymer is a conjugated polymer or a doped conjugated polymer. The conjugated polymer is obtained by chemical oxidative polymerization or oxidative electropolymerization a monomer having a π -conjugated double bond or a derivative thereof. Conductivity is exhibited by adding a small amount of an acceptor that easily accepts electrons or a donor that easily donates electrons to the conjugated polymer. When the acceptor or the donor is added to the conjugated polymer, n electrons are extracted from the conjugated polymer to form negative charge carrier (positive holes, holes) in the case of the acceptor, and electrons are supplied to form negative charge carrier in the case of a donor, and exhibits conductivity.

(44) As conjugated polymers, known polymers can be used without limitation. Examples include polypyrrole, polythiophene, polyfuran, polyaniline, polyacetylene, polyphenylene, polyphenylenevinylene, polyacene, and polythiophenevinylene, and the like. These conjugated polymers may be used alone, or two or more kinds may be combined, furthermore, it may be a copolymer of two or more monomers.

(45) Among the above-mentioned conjugated polymers, a conjugated polymer obtained by polymerizing thiophene or a derivative thereof is preferable, and conjugated polymers polymerized with 3,4-ethylenedioxythiophene (that is, 2,3-dihydrothieno [3,4-*b*][1,4] Dioxin), 3-alkylthiophene, 3-alkoxythiophene, 3-alkyl-4-alkoxythiophene, 3,4-alkylthiophene, 3,4-alkoxythiophene, or derivatives thereof are preferred. Examples of the thiophene derivatives, a compound selected from thiophene with substituents at the 3- and 4-positions is preferable, and the 3- and 4-position substituents of the thiophene ring may form a ring together with the 3- and 4-position carbons. The number of carbons of alkyl and alkoxy groups is suitable to being from 1 to 16, in particular, a polymer of 3,4-ethylenedioxythiophene, called EDOT, i.e., poly(3,4-ethylenedioxythiophene), called PEDOT, is particularly preferred. Further, an alkylated ethylenedioxythiophene in which an alkyl group is added to 3,4-ethylenedioxythiophene may be used, and for example, a methylated ethylenedioxythiophene (that is, 2-methyl-2,3-dihydro-thieno[3,4-*b*][1,4]dioxin), ethylated ethylenedioxythiophene (that is, 2-ethyl-2,3-dihydro-thieno [3,4-*b*][1,4]dioxine), and the like can be mentioned.

(46) As dopant, known dopant can be used without limitation. For example, inorganic acids such as boric acid, nitrate and phosphoric acid, organic acids such as acetic acid, oxalic acid, citric acid, ascorbic acid, tartaric acid, squaric acid, logisonic acid, croconic acid, salicylic acid, p-toluenesulfonic acid, 1,2-dihydroxy-3,5-benzenedisulfonic acid, methanesulfonic acid, trifluoromethanesulfonic acid, borodisalicylic acid, bisoxalate borate acid, sulfonylimide acid, dodecylbenzenesulfonic acid, propylnaphthalenesulfonic acid, butylnaphthalenesulfonic acid can be mentioned. Examples of polyanions include polyvinyl sulfonic acid, polystyrene sulfonic acid, polyallyl sulfonic acid, polyacrylic sulfonic acid, polymethacrylic sulfonic acid, poly (2-acrylamide-2-methylpropanesulfonic acid), polyisoprene sulfonic acid, and polyacrylic acid, polymethacrylic acid, polymaleic acid and the like.

(47) These dopants may be used alone, or two or more kinds may be combined. In addition, these dopants may be polymers of a single monomer or copolymers of two or more monomers. Dopants may also be polymers or monomers.

EXAMPLES

(48) Hereinafter, the electrolytic capacitor of the present invention will be described in more detail based on Examples. The present invention is not limited to the following examples. Various types of electrolytic capacitors were manufactured by coating the sealing member with a resin layer containing an epoxy resin composition.

Examples 1 to 4

(49) The electrolytic capacitors of Comparative Examples 1 and Examples 1 to 4 are common in that they have an electrolyte in which an electrolytic solution and a solid electrolyte are used in combination. The difference between Examples 1 to 4 with respect to Comparative Example 1 is composition of a resin layer, and type of an epoxy resin composition is different. Further, Examples 1 to 4 with respect to Comparative Example 1 are different in that ethylene glycol or γ -butyrolactone is used as a solvent of the electrolytic solution. Ethylene glycol is an example of a compound with hydroxy group.

(50) The common points of the electrolytic capacitors of Comparative Example 1 and Examples 1 to 3 will be described in detail. The anode foil is an aluminum foil, which surface is enlarged by an etching process and dielectric oxide film is formed by a chemical conversion treatment. The cathode foil was a plain foil, i.e., unetched aluminum foil. The same anode foils and cathode foils were used for all electrolytic capacitors, with lead wires connected to each, and the anode foils and cathode foils were wound facing each other through a manila-based separator. The capacitor elements were performed with restorative conversion by immersed in an aqueous solution of ammonium dihydrogen phosphate of the same concentration for 10 minutes.

(51) Next, a dispersion of polyethylene dioxythiophene (PEDOT/PSS) doped with polystyrene sulfonic acid of the same product as a conductive polymer was prepared at the same concentration, the capacitor elements were immersed, the capacitor element were pulled up and dried at 150° C. for 30 minutes. The immersing and drying were repeated the same number of times. As a result, the solid electrolytes were formed on the capacitor elements. Next, a different electrolytic solution was prepared for each electrolytic capacitor, and the capacitor element on which the solid electrolyte was formed was immersed in the electrolytic solution. Each capacitor element was inserted into a bottomed cylindrical outer case of the same metal type, the same size, and the same shape, a sealing member was attached to the end of the opening and sealed by crimping process with the same pressing force. The sealing member made of butadiene rubber was used for each electrolytic capacitor.

(52) Each electrolytic capacitor was housed in an aluminum container of the same size, and a resin layer was formed so as to be completely embedded with the same type of epoxy resin. However, the lead wire drawn from the electrolytic capacitor was drawn from the epoxy resin so that the electrolytic capacitor could be energized. As a result, in each electrolytic capacitor, the outer surface of the sealing member is completely covered with the epoxy resin composition.

(53) The epoxy resin compositions used for the electrolytic solution and the resin layer prepared in the electrolytic capacitors of Comparative Example 1 and Examples 1 to 4 are as shown in Table 1 below.

(54) TABLE-US-00001 TABLE 1 Composition of Electrolytic Solution Ethylene γ - Glycol Butyrolactone (EG) (GBL) Solute Type of Resin Comparative Example 1 100 0 not added acid anhydride-cured epoxy resin Example 1 100 0 not added amine-based epoxy resin Example 2 0 100 not added amine-based epoxy resin Example 3 100 0 not added phenol-based epoxy resin Example 4 0 100 not added phenol-based epoxy resin

(55) As shown in Table 1 above, the electrolytic capacitor of Comparative Example 1 uses an acid anhydride-cured epoxy resin to form the resin layer. On the other hand, the electrolytic capacitor of Examples 1 and 2 use an amine-based cured epoxy resin to form the resin layer. In addition, the electrolytic capacitor of Examples 1 and 4 use phenol-based cured epoxy resin to form the resin layer. Acid anhydride-cured epoxy resin of Comparative Example 1 is the epoxy resin cured by the acid anhydride-based curing agent. The amine-based cured epoxy resin of Examples 1 and 2 is the

epoxy resin cured with the amine-based curing agent. Phenol-based cured epoxy resin of Examples 3 and 3 is the epoxy resin cured with the phenol-based curing agent.

(56) In Table 1 above, solvent composition ratios are weight percentages of total solvents. As shown in Table 1 above, in the electrolytic capacitor of Comparative Example 1, the acid anhydride-cured epoxy resin is used for the resin layer, and the total amount of the solvent in the electrolytic solution is ethylene glycol. In the electrolytic capacitor of Example 1, the total amount of the solvent in the electrolytic solution is ethylene glycol, however the amine-based cured epoxy resin is used for the resin layer. In the electrolytic capacitor of Example 2, the total amount of the solvent in the electrolytic solution is γ -butyrolactone, however the amine-based cured epoxy resin is used for the resin layer. In the electrolytic capacitor of Example 3, the total amount of the solvent in the electrolytic solution is ethylene glycol, however the phenol-based cured epoxy resin is used for the resin layer. In the electrolytic capacitor of Example 4, the total amount of the solvent in the electrolytic solution is γ -butyrolactone, however the phenol-based cured epoxy resin is used for the resin layer.

(57) After filling each electrolytic capacitor with each epoxy resin composition shown Table 1 above, the electrolytic capacitor was left as it is for 400 hours in a temperature environment of 150° C. while applying a constant voltage of 35 V to each electrolytic capacitor. The ESR before and after this heat stress load was measured. ESR was measured at 100 kHz. In addition, the inside of each electrolytic capacitor after being loaded with heat stress was photographed with an X-ray imaging device, and the internal state was observed by a fluoroscopic image.

(58) Table 2 below shows the results of ESR and the internal state of each electrolytic capacitor by fluoroscopic images before and after the heat stress load.

(59) TABLE-US-00002 TABLE 2 Table 2 below shows the results of ESR and the internal state of each electrolyte capacitor by fluoroscopic images before and after the heat stress load. Composition of Electrolytic Solution Ethylene γ - Observation Initial ESR after Glycol Butyrolactone Results with ESR Load Test (EG) (GBL) Solute Type of Resin X-ray Device (Ω) (Ω) Comparative 100 0 not added acid anhydride-cured Cracked 0.066 5.52 Example 1 epoxy resin Example 1 100 0 not added amine-based epoxy resin Unchanged 0.065 0.075 Example 2 0 100 not added amine-based epoxy resin Unchanged 0.085 0.099 Example 3 100 0 not added phenol-based epoxy resin Unchanged 0.066 0.075 Example 4 0 100 not added phenol-based epoxy resin Unchanged 0.085 0.099 Composition of Electrolytic Solution Ethylene γ - Observation Initial ESR after Glycol Butyrolactone Results with ESR Load Test (EG) (GBL) Solute Type of Resin X-ray Device (m Ω) (m Ω) Comparative 100 0 not added acid anhydride-cured Cracked 0.066 5.52 Example 1 epoxy resin Example 1 100 0 not added amine-based epoxy resin Unchanged 0.065 0.075 Example 2 0 100 not added amine-based epoxy resin Unchanged 0.085 0.099 Example 3 100 0 not added phenol-based epoxy resin Unchanged 0.066 0.075 Example 4 0 100 not added phenol-based epoxy resin Unchanged 0.085 0.099

(60) As shown in Table 2, in Examples 1 to 4 with respect to Comparative Example 1, the ESR is an order of magnitude smaller. The groups of Examples 1 to 4 were suppressed to ESR of at least 1/56 and ESR of about 1/74 at maximum, compared to Comparative Example 1. As a result, it was confirmed that when the resin layer is formed with the amine-based cured epoxy resin or the phenol-based cured epoxy resin, ESR after the load test is suppressed regardless of the type of the solvent.

(61) Further, as shown in Table 2, in Examples 1 and 3 in which ethylene glycol was used as the solvent, the increase in ESR was further suppressed as compared with Examples 2 and 4 in which γ -butyrolactone was used as the solvent. When ethylene glycol is present in the solvent while using the solid electrolyte as the electrolyte, the electrical conductivity of the conductive polymer is improved by changing of the higher-order structure of the conductive polymer and reorienting of the crystal structure of the polymer chain, however, since such an effect cannot be obtained with γ -butyrolactone, ESR was further reduced when the solvent was ethylene glycol and the resin layer

was the amine-based cured epoxy resin or the phenol-based cured epoxy resin.

(62) Here, as shown in Table 2, according to the fluoroscopic image taken by the X-ray imaging device, cracks were generated in the resin layer of the acid anhydride-cured epoxy resin in which the electrolytic capacitors of Comparative Example 1 was embedded. On the other hand, no crack could be found in the resin layer of the amine-based cured epoxy resin or the phenol-based cured epoxy resin in which the electrolytic capacitors of Examples 1 to 4 were embedded. FIG. 1 illustrates fluoroscopic images taken from multiple directions of Comparative Example 1, Example 1, and Example 2. As shown in FIG. 1, in Comparative Example 1, cracks are seen in the range surrounded by the dotted line. On the other hand, there are no cracks in each fluoroscopic image of Examples 1 and 2. The area where cracks are observed is between the sealing member made of butadiene rubber and the resin layer of the acid anhydride-cured epoxy resin.

(63) As a result, when the acid anhydride-cured epoxy resin with ester bonds is used for the resin layer and ethylene glycol, which is a compound with hydroxy group, is used as the solvent, it is understood that the acid anhydride-cured epoxy resin is decomposed by hydrolysis, the decomposition products react with the compound with hydroxy group, these reaction products are eluted in the electrolytic solution, and the resin layer-derived components in the electrolytic solution increases ESR. It was confirmed that, by using the amine-based cured epoxy resin or the phenol-based cured epoxy resin without ester bond for the resin layer, such a hydrolysis reaction of the epoxy resin does not occur, the resin layer-derived component does not elute in the electrolytic solution, and the increase in ESR was suppressed.

Examples 5 to 12

(64) Next, the electrolytic capacitors of Examples 5 to 12 were manufactured. In Examples 5 to 12, azelaic acid was added to the electrolytic solution as the anionic component of the solute, and ammonia was added to the electrolytic solution as the cationic component of the solute. The electrolytic capacitors of Examples 5 to 12 have different amounts of cationic components in the solute in the electrolytic solution. The electrolytic capacitors of Examples 5 to 12 were housed in aluminum containers having the same dimensions as Examples 1 to 4, and were completely embedded with the same amine-based cured epoxy resin as Examples 1. In addition, the electrolytic capacitors of Examples 5 to 12 were manufactured by the same manufacturing method and under the same conditions as those of Examples 1 to 4.

(65) After filling each electrolytic capacitor of Examples 5 to 12 with the amine-based cured epoxy resin, the electrolytic capacitor was left as it is for 800 hours in a temperature environment of 150° C. while applying a constant voltage of 35 V to each electrolytic capacitor. The ESR before and after this heat stress load was measured, and rate of change in ESR after heat stress load is calculated. ESR was measured at 100 kHz.

(66) Table 3 below shows the composition and addition amount of the electrolytic solutions of Examples 5 to 12, and ESR (initial ESR) before heat stress load, ESR after heat stress load (post-test ESR), and rate of change in ESR after heat stress load. The amount of the solvent in Table 3 indicates the content with respect to the total solvent, and the amount of the solute in Table 3 is converted per 100 g of the electrolytic solution. The electrolytic capacitors of Examples 5 to 12 are of a winding type with a diameter of 6 mm and a total length of 6 mm, have a rated voltage of 35 V, and have a rated capacity of 47 μ F.

(67) TABLE-US-00003 TABLE 3 Solvent of Electrolytic Amount of cations Solution (Wt %) in 100 g of Initial Post-test Ethylene electrolytic ESR ESR  ESR glycol Sulfolane solution (mmol) (Ω) (Ω) (%) Example 5 40 60 4 0.066 0.078 18 Example 6 17 0.066 0.096 45 Example 7 25 0.073 0.113 56 Example 8 34 0.069 0.142 105 Example 9 42 0.071 0.209 194 Example 10 51 0.071 0.292 312 Example 11 76 0.071 0.982 1284 Example 12 86 0.071 1.741 2353

(68) As shown in Table 3, the change rate of ESR of the electrolytic capacitor of Example 11 having 76 mmol of ammonia was suppressed to about 55% with respect to the electrolytic capacitor of Example 12 having 86 mmol of ammonia as the cationic component. Furthermore, the change

rate of ESR of the electrolytic capacitor of Example 10 having 51 mmol of ammonia was suppressed to about 24% with respect to the electrolytic capacitor of Example 11 having 76 mmol of ammonia. Furthermore, the change rate of ESR of the electrolytic capacitor of Example 7 having 25 mmol of ammonia was suppressed to about 18% with respect to the electrolytic capacitor of Example 10 having 51 mmol of ammonia.

(69) As described above, when the electrolytic capacitor has the resin layer, the cationic component tends to promote the dedoping reaction and increase the ESR. However, it was confirmed that the increase in ESR can be suppressed by adjusting the cationic component to 76 mmol or less per 100 g of the electrolytic solution, and the increase in ESR can be further suppressed by adjusting the cationic component to 51 mmol or less per 100 g of the electrolytic solution, ESR after heat stress load can be greatly suppressed by adjusting the cationic component to 25 mmol or less per 100 g of the electrolytic solution.

Examples 13 to 16

(70) The electrolytic capacitors of Examples 13 to 16 were manufactured. The solvent of the electrolytic solution of Examples 13 to 16 is ethylene glycol, γ -butyrolactone, sulfolane, or a mixed solution of two of these, however the composition ratios are different. In the electrolytic capacitors of Examples 13 to 16, no solute was added from the viewpoint of clarifying the influence of only the solvent of the electrolytic solution. In addition, the electrolytic capacitors of Examples 13 to 16 were housed in aluminum containers of the same size and completely embedded with the resin, and were manufactured by the same manufacturing method and under the same conditions as those of Examples 1 to 4. The amine-based cured epoxy resin was used for the resin layer.

(71) After filling with the resin, each electrolytic capacitor was left as it is for 3500 hours in a temperature environment of 150° C. The released amount of electrolytic solution after this heat stress load was measured. The released amount of the electrolytic solution was obtained by subtracting the weight of the electrolytic capacitor after the heat stress load from the weight of the electrolytic capacitor before the heat stress load. The composition and the adding amount of the electrolytic solution in Examples 13 to 16 and the released amount of the electrolytic solution after heat stress load are shown in Table 4 below. The electrolytic capacitor of each Example is of a winding type with a diameter of 10 mm and a total length of 10 mm.

(72) TABLE-US-00004

TABLE 4 Solvent Ratio of Electrolytic Solution (Wt %)	Released Amount
Ethylene γ - of Electrolytic glycol Butyrolactone Sulfolane Solution (mg)	Example 13 0 100 0 -125
Example 14 0 0 100 -50	Example 15 40 0 60 -49
Example 16 100 0 0 -37	

(73) As shown in Table 4, compared to Example 13 in which γ -butyrolactone was used as the solvent of the electrolytic solution, the released amount of the electrolytic solution in Example 14 in which sulfolane was used as the solvent of the electrolytic solution was reduced to less than half. In addition, compared to Example 14 in which sulfolane was used as the solvent of the electrolytic solution, the released amount of the electrolytic solution in Example 16 in which ethylene glycol was used as the solvent of the electrolytic solution was reduced to 74%.

Claims

1. An electrolytic capacitor, comprising: a capacitor element including an anode foil, a cathode foil, a solid electrolyte, and electrolytic solution, a case housing the capacitor element, a sealing member sealing the case, and a resin layer arranged in the vicinity of the sealing member, wherein: the resin layer includes an epoxy resin composition made from a phenol-based curing agent or an amine-based curing agent and an epoxy resin as raw materials without ester bond, and cationic component in the electrolytic solution is contained in an amount of 34 mmol or less per 100 g of the electrolytic solution.

2. The electrolytic capacitor according to claim 1, wherein the electrolytic solution contains at least one of a compound with hydroxy group, sulfolane and γ -butyrolactone.

3. The electrolytic capacitor according to claim 2, wherein the compound with hydroxy group is at least one of ethylene glycol, diethylene glycol, or polyethylene glycol.
 4. The electrolytic capacitor according to claim 1, wherein the cationic component in the electrolytic solution is contained in an amount of 25 mmol or less per 100 g of the electrolytic solution.
 5. An electrolytic capacitor, comprising: a capacitor element including an anode foil, a cathode foil, a solid electrolyte, and electrolytic solution, a case housing the capacitor element, a sealing member sealing the case, and a resin layer arranged in the vicinity of the sealing member, wherein: the resin layer includes an epoxy resin composition without ester bond, wherein: the epoxy resin composition has a chemical structure containing a phenol-based curing agent or an amine-based curing agent and an epoxy resin, and cationic component in the electrolytic solution is contained in an amount of 34 mmol or less per 100 g of the electrolytic solution.
 6. The electrolytic capacitor according to claim 3, wherein the electrolytic solution contains at least one of a compound with hydroxy group, sulfolane and γ -butyrolactone.
 7. The electrolytic capacitor according to claim 6, wherein the compound with hydroxy group is at least one of ethylene glycol, diethylene glycol, or polyethylene glycol.
 8. The electrolytic capacitor according to claim 5, wherein the cationic component in the electrolytic solution is contained in an amount of 25 mmol or less per 100 g of the electrolytic solution.
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